

STAR PXL Single-Track Pointing Resolution Estimate

Detector inputs

- Two pixel layers at radii

$$r_1 = 2.8 \text{ cm}, \quad r_2 = 8.0 \text{ cm}.$$

- Pixel pitch: $20.7 \mu\text{m} \times 20.7 \mu\text{m}$, so the single-hit resolution is

$$\sigma_{\text{hit}} = \frac{20.7}{\sqrt{12}} = 5.98 \mu\text{m}.$$

- Pixel material: $0.52\% X_0$ per layer.
- Beam pipe: $r_{\text{BP}} = 2.0 \text{ cm}$, material $0.24\% X_0$.

Hit-resolution limit

- For a straight track $y(r) = a + br$, the intercept at the collision point is

$$a = \frac{r_2 y_1 - r_1 y_2}{r_2 - r_1}.$$

- Propagating the two independent hit errors gives

$$\sigma_{a,\text{hit}} = \sigma_{\text{hit}} \sqrt{\left(\frac{r_2}{r_2 - r_1}\right)^2 + \left(\frac{r_1}{r_2 - r_1}\right)^2} = 9.74 \mu\text{m}.$$

Multiple-scattering term

- Using the Highland approximation for $\beta \simeq 1$,

$$\theta_0 \simeq \frac{13.6 \text{ MeV}}{p} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x}{X_0} \right) \right].$$

- The beam pipe contributes

$$2.0 \text{ cm} \times \frac{0.5135 \text{ mrad}}{p} = \frac{10.27 \mu\text{m GeV}/c}{p}.$$

- The first PXL layer contributes

$$2.8 \text{ cm} \times \frac{0.7847 \text{ mrad}}{p} = \frac{21.97 \mu\text{m GeV}/c}{p}.$$

- The outer PXL layer is the last measured point in this two-hit estimate, so its post-hit scattering does not affect the fitted pointing.

Final fast estimate

$$\sigma_{\text{point}}(p) \simeq \sqrt{(9.74 \mu\text{m})^2 + \left(\frac{24.25 \mu\text{m GeV}/c}{p} \right)^2}, \quad p \text{ in GeV}/c.$$

$p \text{ [GeV}/c]$	$\sigma_{\text{point}} \text{ [}\mu\text{m]}$
0.2	121.7
0.5	49.5
1.0	26.1
2.0	15.6
5.0	10.9
10.0	10.0

This estimate is for central tracks and uses the same resolution in $r\phi$ and z .